CURRENT FILE

Bt102

Effort by: MAAH WARRIORS + ASHAM

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Linnaeus (in 1735) established the system of scientific nomenclature. Each organism has two names: the genus and specific epithet (descriptive word).
Are italicized or underlined. The genus is capitalized, and the specific epithet is lowercase. Escherichia coli

Aristotle: stated that living organisms could arise from nonliving matter and that such a process was regular in nature. For example dust creates fleas, bread or wheat left in a dark corner produces mice

Robert Koch in 1876 purposed the germ theory.
Germ theory states that specific microscopic organisms are the cause of specific diseases.

- Deals with diseases of humans and animals
- Identifications of disease causing agents
- Control and elimination of microbes
- Tracking down new pathogens

Eyepiece, Eye tube, Objective lenses, Nosepiece, Adjustment knobs, Stage, Illuminator, Condenser and condenser focus knob, Diaphragm

- Darkfield microscopy
- Phase contrast microscopy
- Fluorescent microscopy
- Transmission electron microscopy
- Scanning electron microscopy

- Applying primary stain (crystal violet).
- Adding mordant (Gram's iodine).
- De-colorization with ethanol, acetone or a mixture of both.
- Counterstaining with safranin.

Some bacteria such as Mycobacteria have a waxy material in their cell wall. Specific name for this waxy substance is mycolic acid

Principle of Acid-Fast Staining:
Mycobacteria are lipophilic, not easy to stain but once stained, they are resistant to acidic alcohol decolorization process.

Procedure for Acid-Fast Staining:

- Make a smear
- Carbolfuchsin on slides
- Heat the slide for 5-10 min
- Wash with water and then with acid alcohol for 1 min.
- Counterstain with methylene blue for 1 min
- See under the microscope.
- Acid fast bacteria have mycolic acid that can be stained by heating the organisms with the stain

9. Difference between prokaryotes and eukaryotes 12

PROKARYOTES	EUKARYOTES
Prokaryotes are unicellular	Eukaryotes are multi-celled organisms
DNA is stored in the cytoplasm of prokaryotes.	DNA is stored within the nucleus of eukaryotes
Nucleus is absent	Nucleus is present
Membrane bound organelles is absent	Membrane organelles is present
One chromosome is present, but not true	More than one number chromosomes present
Examples: bacteria and archaea	Examples: protist, plants, fungi, animal

10.Shapes of bacteria 13

Three types:

- **Coccus:** These are circular or spherical in shape.
- **Bacillus:** These rod shaped bacteria.
- **Spiral:** These are curved shaped bacteria. They are further divided into 3 more subgroups.
  - **Vibrio:** curved rods
  - **Spirillum:** Helical but rigid
  - **Spirochete:** Helical but flexible Move by axial filaments

11.Fimbriae location & function 16

- Hair-like small appendages on G-neg cells
- Composed of pilin, a protein
- Can be at one pole or around the entire cell
- Used for attachment to surfaces of epithelial cell
- Fimbriae help bacteria attach to surfaces
- Fimbriae also help attach to epithelial cells e.g. Neisseria gonorrhoeae

12.Three components of lipopolysaccharides? 18

- Lipid A: endotoxin – Fever, vasodilation and shock
- Core polysaccharide: Structural support
- O side chain: Antigenic like teichoic acid in gram positive bacteria

13.Function of ribosomes, cell wall and flagella 22

- **Cell wall:** gives support and protection against plasmolysis.
- **Flagella:** help bacteria in movement.
- **Ribosomes:** are organelles that assist in protein synthesis during translation.

14.Define metachromatic granules 22

Metachromatic granules are large inclusion that used for store of inorganic phosphate. It is found in the bacteria. It is also called volutin. It appears red color with methylene blue.

15.Germination of spore 23

Germination has 3 phases or stages:

- **Activation:** It prepares the spore for germination.
- **Germination:** Spore starts swelling and losing its coats etc.
- **Outgrowth:** New components are made.

16.Define anaerobic respiration. And three anaerobic bacteria 25

Anaerobic Respiration: If in the electron transport chain, the final electron acceptor is any inorganic compound other than oxygen, it is called an anaerobic respiration. This final electron acceptor may be a nitrate ion, a sulfate ion, or a carbonate ion.

Bacteroides, Prevotella, Actinomyces, Clostridia

17.Three stage of respiration of glucose 25

- **Glycolysis:** it is the oxidation of glucose to pyruvic acid with ATP and NADH.
- **The Krebs cycle:** Oxidation of acetyl CoA to carbon dioxide with ATP, NADH and FADH2
- **Electron transport chain:** NADH and FADH2 are oxidized to produce ATP

18.Bacterial classes in on the basis of temperature 27

- **Psychrophiles:** These are further divided into strict psychrophiles and psychrotrophs:
  - **Psychrophiles:** Cold loving: 15°C.
  - **Psychrotrophs:** Optimum temp 1s 20-30°C.
- **Mesophiles:** Moderate temp loving 25 – 40°C, Optimum 37°C
- **Thermophiles:** Heat loving 50 – 60°C

19.Enlist four phases of bacterial growth curve 33

- **The Lag Phase:** initial period of bacterial cells in new environment before starting growth.
- **The Log Phase:** in this phase bacteria growth occur and cell number increase
- **The Stationary Phase:** In the stationary phase, growth reaches a point in which number of dying cells equals the number of dividing cells. This is also called a period of equilibrium
- **The Death Phase:** The number of death cells increase and number of new cells decreased in this phase is called death phase.

20.Which factors can influence the bacterial death rate 33

The reason is simple: Nutrients are depleted and waste products which are toxic to the cells accumulate suppressing the growth and killing the cells.

The rate of growth or death of a particular microbial species is influenced by a variety of physical factors in its environment including temperature, osmotic pressure, pH, and oxygen concentration.

21.Uses of growth curves 33

Microbial growth curves are used to study differential effects of media, genetics, and stress on microbial population growth. Also use for:

- Vaccine production
- Seed for bacteria

- Culture preservation

22.Sepsis, antisepsis , sanitization 35

- **Sepsis:** Sepsis is the extreme response of body to an infection
- **Antisepsis:** Removing pathogens from living tissue
- **Sanitization:** removal bacteria from surface

23.Plasma membrane disrupt 40

Hexacholorphene, triclosan, Chlorhexidine, Alexidine

24.State mode of diseases cause with higher pressure and osmotic pressure? 42

- High pressure deshapes and denatures proteins: This technique is used for preserving juices and wines.
- Osmotic pressure: causes plasmolysis. High salt concentrations

25.Dry heat method? 42

- Direct flaming
- Incineration
- Hot air sterilization(hot-air oven)

26.Define, Gene, Genome, Chromosomes and phenotypes 44

Gene: The basic unit of heredity passed from parent to child
Genome: The genome is the entire set of DNA found in a cell.
Chromosome: threadlike structures made up of protein and DNA that carry genomic information from cell to cell.
Phenotype: all observable characters that influenced by genotypes and environment

27.Classification of five kingdom system 48

In 1969 the five kingdom classification was proposed by Robert whittakar

- **Monera**: bacteria and archaea
- **Plantae:** plants
- **Animalia :** Animals
- **Protista:** they are unicellular eukaryotes.
- **Fungi:** yeast mold mushrooms, consist of chitin

28.Detail note on western blotting method 53

We can use this technique for the detection of antigen in the serum. Microbial protems can be separated on SDS-PAGE by electrophoresis and the presence of these proteins can be detected by enzyme-tagged antibodies specific to those proteins. A color band will be seen where the specific protein (antigen) 1s present on the gel. Please remember that proteins in the gel are first transferred to a paper strip before they could be detected by specific antibodies

29.Steps of preparing stain of western bloating? 53

- Proteins from microbes are separated by electrophoresis
- Protein bands are blotted onto a nitrocellulose membrane.
- Proteins could be detected by electrophoresis and blotting.
- Blots can be detected by specific antibodies labeled with radioisotopes etc.

30.Probe & Its use in biotechnology 55

A probe is a single-stranded sequence of DNA or RNA used to search for its complementary sequence in a sample genome. And its uses:

- DNA probes can be used for diagnosing various diseases.

- Rapid detection and identification of infectious agents
- Use in PCR, qPCR and hybridization techniques for detection
- Ensure high specificity and sensitivity
- Detection of nonviable organism

31.PCR denaturing annealing and extended stage note 56

PCR goes through 3 step cycles for about 30 to 35 times. Each cycle consists of:

- **Denaturing stage (about 95oC):** when the double-stranded template DNA is heated to separate it into two single strands.
- **Annealing (50 to 56oC):** in which step temperature is lower and DNA primers attach to DNA template
- **Extending stage (72oC):** when the temperature is raised and the new strand of DNA is made by the Taq polymerase enzyme.

32.Hyphae, thallus, mycelium 56

- **Hyphae:** Thread like filaments of a fungus
- **Mycelium:** Collection of hyphae
- **Thallus:** Collection of mycelia

33.Life cycle of fungi 57

Filamentous fungi can reproduce asexually by fragmentation of their hyphae. This fragmentation takes various names according to the species of the fungus. Spores can also be produced sexually by these fungi; however, this happens only when food is scarce and environmental conditions are not favorable for further growth of filamentous fungi.

Asexual spores type:

- **Conidia:** Chain of spores at the end of a conidiphore Aspergillus
- **Arthroconidia:** spores formed by fragmentation of septate hypha
- **Blastoconidia:** spores as budding of the parent cells
- **Chlamydioconidia:** spores formed by rounding and enlargement within a hyphal segment

34.Sporangiospore and chlamydioconidia 57

Sporangiospores: Formed within a sporangium or sac at the end of an aerial hypha called sporangiophore

Chlamydioconidia: spores formed by rounding and enlargement within a hyphal segment

35.Sexual reproduction in fungi 58

Sexual Spores

- **Plasmogamy:** A haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-).
- **Karyogamy:** The (+) and (-) nuclei fuse to form a diploid zygote nucleus.
- **Meiosis:** The diploid nucleus gives rise to haploid nuclei (sexual spores)

36.Basic features of yeast 59

- Non-filamentous fungi
- Unicellular Reproduce by budding (asexually) or sexually
- Some are dimorphic
- Larger than bacteria
- No flagella or cilia
- Facultative anaerobes

- Gram positive

37.Importance of fungi 61

Saccharomyces are used for alcoholic beverages under anaerobic conditions for alcohol production (fermentatively). However, if Saccharomyces are incubated aerobically (for bread making), they metabolize glucose to produce CO2 (some ethanol is also produced but gets evaporated during baking) which expands the dough causing it to rise. This yeast is also extensively used for molecular biology work and vaccine production. Vitamin C is obtained from Aspergillus niger; cellulase from Trichoderma, antibiotics from many fungi and anticancer Taxol from Taxomyces are some other examples of useful fungal products. Entomophaga (kills gypsy moth), hence is used as a biological pest control. Mushrooms can be cooked and eaten as a source of proteins for humans.

38.Algae & its structure 63

The definition of algae is a single or multi-cellular organism that has no roots, stems or leaves and is often found in water. These are simple eukaryotic cells.

Structure: Body of multicellular alga such as seaweed is called a thallus which consists of branched holdfasts (anchor alga to rock) stem like hollow stipes and leaf like blades. There is no vascular tissue in these algae. Surrounding water provides the support for the thallus. Some algae have a gas filled body inside them which keeps them floating in the water. This gas filled structure is called pneumatocyst or float

39.Phaeophyta, Rhodophyta, Dinoflagellates, Chlorohyta, Bacillophyta 64

Phaeophyta (Kelp): Commonly known as brown algae, brownish in color, contains cellulose and alginic acid cell walls, are multicellular and contain chlorophyll a and c, xanthophylls. These store carbohydrates and are used for algin which is a thickener in ice creams. Can grow upto 20 cm in one day

Rhodophyta: Red Alage, these are reddish in color and contain cellulose in their cell walls. Most are multicellular and contain chlorophyll a and d, and phycobiliproteins. They store glucose polymer and used for agar and carrageenan

Chlorophyta: Green Algae: have all the features of red algae except that chlorophyll is of a and b types only. These algae are believed to have given rise to terrestrial plants.

Bacillariophyta: Cell wall is composed of pectin and silica. They are unicellular and possess chlorophyll a and c, and carotene xanthophylls. These algae store oil and produce domoic acid with which human can be intoxicated by eating contaminated mussels

Dinoflagellata: These are unicellular algae collectively called as plankton. These cells have cellulose in their cytoplasmic membrane which gives the membrane rigidity. They have chlorophyll a and c, and carotene. These algae store starch in them.

40.Consequences of viral infections 68

- Lysis of cells
- Microscopic and macroscopic Degenerative changes (Cytopathic Effects)
- **Virulent infection:** lysis of cells
- **Latent infection:** Virus does not replicate
- **Persistent infection:** Virus replicates but slowly
- **Transformation:** Tumor cells

41.Define epidemiology and its three methods 71

The study of where and when diseases occur and how they are transmitted in a population is called epidemiology. Methods are

- **Descriptive:** (recording the data about disease)
- **Analytical:** (To determine the cause of disease)
- **Experimental Epidemiology:** Hypothesis of a disease is tested
- **Case reporting:** Health care workers report specified disease to local, state, and national offices

42.Morbidity, mortality, Pathogenicity 71

- **Morbidity:** Incidence of a disease
- **Mortality:** Deaths from a disease
- **Pathogenicity:** The ability to cause disease

43.Microbiota Factor And Distribution Composition 72

Availability of various nutrients determines the type of organisms present in a system. Similarly, physical and chemical factors also influence the type of microbiota. Other factors include pH, O2, and CO2.

44.Endemic, epidemics, pandemic 73

Endemic: When a disease is consistently present, but with low incidence, we call it an endemic disease. Common cold is a typical example.

Epidemic: Within an area, unusually high incidence of a disease is called an epidemic. Influenza is a good example.

Pandemic: When a disease assumes a global occurrence or becomes worldwide, it is called an epidemic. AIDS is a good example of such as epidemic.

45.Types of Transmission of disease 74

- **Direct Transmission:** Person to person physical contact is a direct contact. Influenza spread is a typical example.
- **Indirect Transmission:** A nonliving object is involved. Objects in the use of a patient such as handkerchief, utensils, pillow and bedding are called fomites. These can transmit organisms to susceptible individuals.
- **Droplet Transmission:** Mucus droplets are created when you sneeze for example. These droplets carry organisms that may infect another individual who comes in contact with the droplets. One sneeze may produce 20,000 droplets. Influenza can be spread by this route.
- **Vehicle Transmission:** Transmission of disease agents by a medium, such as water, food, or air. Other media include blood and other body fluids, drugs, and intravenous fluids
- **Vectors:** A vector is a vehicle (animate or inanimate) that carries a pathogen from one host to another. Arthropods are vectors for many diseases. There are two ways these arthropods can transmit organisms to susceptible hosts:
  - **Mechanical Transmission:** In this type of transmission, organisms stick to the body parts of insects and get transmitted passively from one location to another.
  - **Biological Transmission:** Microbes multiply in the insect and is transmitted actively by insect bites from one individual to another

46.Explain natural and acquired immunity and active, passive immunity 78

Ability of the body to protect itself from harmful effects of diseases

Two types:

- **Innate Immunity (natural):**

Plants and animals have what is called mnate immunity. Innate immunity is the first line of defense against pathogens. It involves several cell types, proteins, and even an organ. The organ involved is your skin. Yes, skin is part of the first line of defense. It protects you and prevents pathogens from getting inside your body.

Naturally Acquired Active Immunity: This type of adaptive immunity develops in a person who gets infected with a microbe but then recovers from that illness making antibodies.

Naturally Acquired Passive Immunity: A typical example of this type of immunity is transfer of antibodies from colostrum or milk of a mother to the offspring.

- **Adaptive immunity (acquired):**

Adaptive immunity is an important part of the immune system. It is protection from an infectious disease agent that is mediated by B- and T- lymphocytes following exposure to specific antigen, and characterized by immunological memory

Artificially Acquired Active Immunity: When a person receives a vaccine against a microbe and the body makes antibodies in response to that vaccine, it is called artificial acquired active immunity because body makes these antibodies actively by itself.

Artificially Acquired Passive Immunity: When premade antibodies are given to a person in order to protect a person from an illness or a snake bite, this constitutes an artificially acquired passive immunity type. The following figure illustrates these four types

47.What is PAMPS? 79

Pathogen associated molecular patterns (PAMPs): Formed by LPS, flagellin, peptidoglycan, DNA, RNA

48.Inflammation and its consequence? 80

Damage to body tissues triggers a local defensive response which is called inflammation. It causes:

- Redness
- Pain
- Swelling

Heat During inflammation, various mediators of inflammation is released, some of which include:

- Histamine
- Kinins
- Prostaglandins by damaged cells
- Leukotrienes by mast cells
- Cytokines are released (TNF-alpha) o Cause's release of acute phase proteins from liver.

49.Interferon and it's types 80

These are antiviral proteins, produced by macrophages and lymphocytes in response to viral infections. Fibroblasts can also secrete it. These proteins interfere with viral replication.

Species specific: This means that interferon produced by humans cannot be used in dogs or cats or vice versa.

Not viral specific: This means that interferon produced in response to virus A can also be effective against virus B.

Its types:

- **Alpha:** It is released by all infected cells.

- **Beta:** Also produced by infected cells.
- **Gamma:** This one is released by lymphocytes only. Lymphocytes belong to adaptive immune system (just remember that).

50.Antigen and epitope 82

Antigen: A foreign substance that provokes immunity is called an antigen. The definition of antigen is a harmful substance which enters the body which causes the body to make antibodies as a response to fight off disease. More specifically called as an immunogen. Proteins are best antigen, though glycoproteins and lipoproteins can also act as antigens. Carbohydrates and lipid alone are poor antigens.

Epitopes: Specific regions on an antigen. In other words, any antigen has multiple regions each one of those acts as a small antigen. These are called epitopes. These epitopes are also called as antigenic determinants. Most antigens have MW >10,000 D. Molecules that are less than 10,000 D are generally poor antigen.

51.Clonal selection theory 83

According to this theory, B cells are generated in the bone marrow randomly without seeing the antigen. When such a B cell encounters its specific antigen in the body, it starts proliferating into a clone which ultimately becomes plasma cells and start secreting antibodies. A beautiful example to understand clonal selection theory is like going to a shoe store and selects a shoe size that fits you very well. Note that the shoe that you selected was made randomly without any regard to your foot size. Clonal selection theory is exactly like that. Antigen selects the clone which fits the antigen.

52.Name 5 antigen presenting cells 85

These cells are able to present antigens to B cells for making antibodies by B cells. Three cells are: Dendritic cells, macrophages, and B cells

53.Primary & secondary immune response 86

When a person is exposed to an antigen (a microbe for example), the antigen is presented to B cells which start making antibodies. Initially, there are no detectable antibodies in the serum. However, after 3-5 days, IgM antibodies are secreted and then 7-10 days after, the same B cell starts making IgG. The level of IgG rises to a level and then it starts declining. This is a typical response of an antigen exposure and is called a primary immune response.

It is interesting to note that a second exposure with the same antigen results in a heightened response in terms of very high levels of antibodies that are achieved in a very short time (within 2-7 days). This is called a secondary immune response. It is also called a memory or anamnestic response because memory cells are involved that ensure a quick response to the second exposure by the same antigen

54.Hypersensitivity and types 92

It is a normal Immune response but in a damaging way. In other words, immune system is acting in a way for which it has made to. So, when an antigen gets into the body, immune system recognizes it as foreign and start making antibodies against the antigen. Allergy is another name for hypersensitivity. Those antigens that act in a bad way are called allergens

Types:

- **Type I (Anaphylactic):** Also called anaphylaxis or immediate hypersensitivity. This is IgE mediated. Happens in<30 mints
- **Type II (Cytotoxic):** IgG and IgM mediated as these antibodies attach to cells and destroy them by activating complement

- **Type III (Immune complex mediated):** Immune complexes are formed by IgM or IgG antibodies and these complexes lodge in the capillary beds and cause damage by complement activation.
- **Type IV (delayed type cell-mediated):** CD4+ helper cells along with macrophages are involved.

55.Anaphylaxis ,causes, diagnosis & treatment 92

Anaphylaxis is a severe, potentially life-threatening allergic reaction. It can occur within seconds or minutes of exposure to something you're allergic to, such as peanuts or bee stings

- **Causes:** The patient also experiences difficulty in breathing. It is an emergency as it may kill the person. Anaphylaxis may be local or systemic. Shock from drug reactions, venoms, and common allergens causing asthma are some examples of type I hypersensitivity.
- **Diagnosed:** Diagnosed by demonstration of IgE by intradermal injections of allergens one by one. If specific IgE are present in the serum, a reaction will be visible on the skin locally in the form of a wheel of inflammation.
- **Treatment:** Epinephrine & antihistaminic drugs provide a means of treatment.

56.Type 2 hypersensitivity diagnosis and treatment 93

This hypersensitivity involves antigenic cells and IgG or IgM antibodies that become reactive with these body cells. Whenever, IgM or IgG antibodies bind to their antigens, they always activate complement which then can punch holes in the cells lysing them altogether. Blood transfusion reaction is a good example of type II hypersensitivity.

Diagnosis: Cell deficiencies, Low number of RBCs due to hemolysis, Thrombocytopenia if platelets are involved.

Treatment: Avoid those drugs that cause it. Blood transfusion with correct blood type, Symptomatic treatment, treated according to the cause

57.Antibiotics 97

Antibiotics are produced by microbes, but only a few are selectively toxic to microbes. Antibiotics are products of microbes against microbes. Must not be toxic to the body

58.Inhibits Growth Cell Wall Drug Name? 98

Penicillin's and cephalosporins are the major antibiotics that inhibit bacterial cell wall synthesis.

59.Antivirus drugs for nucleotides and nucleoside and uses 102

- Acyclovir
- Ganciclovir
- Ribavirin

Used mostly for herpesviruses

60.Disc diffusion method

The method consists of placing paper disks saturated with antimicrobial agents on a lawn of bacteria seeded on the surface of an agar medium, incubating the plate overnight, and measuring the presence or absence of a zone of inhibition around the disks

61.Microbes enter the body through which source

Microbes enter the body through ingestion, inhalation, direct contact, cutaneous penetration, and ascending infection

62.How antibodies are effective against microbes

Antibodies help the body to fight microbes or the toxins (poisons) they produce. They do this by recognising substances called antigens on the surface of the microbe, or in the chemicals they

produce, which mark the microbe or toxin as being foreign. The antibodies then mark these antigens for destruction. Antibodies have a long, successful and yet bumpy history of effectiveness against viruses and bacteria.

63.How microbes become resistant to antimicrobial

Antimicrobial resistance is accelerated when the presence of antibiotics and antifungals pressure bacteria and fungi to adapt. Antibiotics and antifungals kill some germs that cause infections, but they also kill helpful germs that protect our body from infection. The antimicrobialresistant germs survive and multiply

64.Physical & Chemical method used to fix smear

Chemical fixation is the most common approach in specimen preservation. Tissues are immersed in a fixative that kills and stabilizes the cell contents. Physical fixation can include microwaving and cryopreserving samples to rapidly inactivate cellular activity